Grand Rosetta Table: Complete Classification of Polynomial Continued Fraction Families and the Dichotomy Principle

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Abstract

We present a complete classification of four polynomial continued fraction (PCF) families discovered via a systematic search over quadratic–linear parameter spaces. All families share the structure $a_n = (k_0 + 2m\sigma)\, n - 2n^2$, $b_n = \varepsilon + 3n$, and obey a universal Gamma-function ratio formula $V(m)/V(m{-}1) = (m+\alpha)/(m+\beta)$. A sharp dichotomy emerges: when $\alpha - \beta \in \mathbb{Z}$ the Pochhammer symbol telescopes to a polynomial, yielding purely rational values (*T-type*); when $\alpha - \beta \in \mathbb{Z} + \frac{1}{2}$ the values are transcendental shifts of the classical S-family involving $\sqrt{\pi}$ and the Gamma function. All identities are verified to 550–2000 digits of precision via high-precision backward evaluation and PSLQ integer-relation detection.

1 Introduction

A *polynomial continued fraction* (PCF) is an expression of the form

$$V \;=\; b_0 + \cfrac{a_1}{b_1 + \cfrac{a_2}{b_2 + \cfrac{a_3}{\ddots}}} \tag{1}$$

where a_n and b_n are polynomials in n. When the limit exists, it often equals a closed-form combination of classical constants. The programme of systematically discovering such identities was initiated by the Ramanujan Machine project [1], which used massively parallel search to rediscover and extend formulas for π, e, and $\zeta(3)$. Theoretical foundations connecting PCFs to hypergeometric functions are developed in [3, 2], while the classical relationship between continued fractions and Padé approximants dates to Euler and Gauss [4].

The *S-family*, defined by $a_n = -n(2n - 2m - 1)$ and $b_n = 3n + 1$, is a one-parameter family whose $m = 0$ member yields the well-known $S^{(0)} = 2/\pi$ (equivalent, via Euler's continued fraction theorem, to the Wallis product [5]). Its closed form is given by the Gauss summation of a ${}_2F_1$ hypergeometric series evaluated at unit argument:

$$S^{(m)} \;=\; \frac{2\,\Gamma(m+1)}{\sqrt{\pi}\,\Gamma(m+\frac{1}{2})}\,. \tag{2}$$

This identity, together with the inter-member ratio $S^{(m)}/S^{(m-1)} = 2m/(2m-1)$, serves as the "Rosetta Stone" template for our search: we scan all quadratic a_n, linear b_n families over small integer coefficients and flag those whose consecutive members have exact rational ratios, then classify the survivors.

In this paper we report the results of this systematic scan. All four families found are completely classified in the Grand Rosetta Table (Table 1), and a sharp *Dichotomy Principle* (Theorem 2) is established distinguishing rational from transcendental families by the arithmetic nature of a single parameter.

2 Setup and family parameterisation

We consider one-parameter families of PCFs of the form (1) with

$$a_n(m) = (k_0 + 2m\sigma)\, n - 2n^2, \qquad b_n = \varepsilon + 3n, \qquad b_0 \in \{1, 2\}, \tag{3}$$

where $k_0 \in \mathbb{Z}$ is a constant offset, $\sigma \in \{+1, -1\}$ controls the sign of the m-coupling, and $\varepsilon \in \{1, 2\}$ is the b-intercept. The parameter m ranges over the non-negative integers (and analytically continues to half-integers and beyond).

The systematic search scans all $(k_0, \sigma, \varepsilon, b_0)$ with $|k_0| \leq 5$, evaluates each family at $m = 0, 1, \ldots, 5$ to 60-digit precision via backward continued-fraction evaluation (depth $N = 5000$), and applies a *Wallis check*: the consecutive ratios $V(m)/V(m-1)$ are tested for exact rationality by searching for p/q with $q \leq 200$ matching to > 50 digits. Families passing this filter are promoted to full verification at 550–2000 digits.

The four families that survive are listed below with their explicit parameters:

Family	k_0	σ	ε	b_0	$a_n(m)$	b_n
1	-3	-1	2	2	$-(2m{+}3)\, n - 2n^2$	$3n + 2$
2	-1	-1	2	2	$-(2m{+}1)\, n - 2n^2$	$3n + 2$
3	$+3$	$+1$	1	1	$(2m{+}3)\, n - 2n^2$	$3n + 1$
4	$+4$	$+1$	1	1	$(2m{+}4)\, n - 2n^2$	$3n + 1$

Families 1–2 have $b_0 = 2$ and $\sigma = -1$ (negative m-coupling); Families 3–4 have $b_0 = 1$ and $\sigma = +1$ (positive m-coupling). This structural split is the origin of the dichotomy.

3 The Grand Rosetta Table

For each family, the consecutive-member ratio takes the universal form

$$\frac{V(m)}{V(m-1)} = \frac{m + \alpha}{m + \beta} \tag{4}$$

for specific constants α, β. Iterating this recurrence from the base value $V(0)$ gives the universal closed form

$$V(m) = V(0)\, \frac{\Gamma(m + \alpha + 1)\, \Gamma(\beta + 1)}{\Gamma(\alpha + 1)\, \Gamma(m + \beta + 1)}\,. \tag{5}$$

Table 1 collects the parameters α, β, the base value $V(0)$, and the closed form for all four families. Every entry has been verified to at least 550 stable digits by comparing the PCF value (backward evaluation at depth $N = 5000$, precision 600 dp) against the right-hand side of (5).

Remark 1 (Base values). Family 3 satisfies $V_3(0) = S^{(1)} = 4/\pi$, which follows from the Wallis-product representation of the S-family at $m = 1$ [5]. Family 4 satisfies $V_4(0) = S^{(3/2)} = 3/2$; here the Gamma functions in (2) evaluate to $\Gamma(5/2) = 3\sqrt{\pi}/4$ and $\Gamma(2) = 1$, giving $2 \cdot (3\sqrt{\pi}/4)/(\sqrt{\pi} \cdot 1) = 3/2$.

Table 1: Complete classification of Rosetta Stone PCF families. The telescoping condition $\alpha - \beta \in \mathbb{Z}$ causes all transcendental factors to cancel, yielding rational closed forms.

Fam	Label	α	β	$\alpha-\beta$	$V(0)$	Ratio	Closed Form	Type
1	$T^{(m+1)}$	$\frac{3}{2}$	$\frac{1}{2}$	1	3	$\dfrac{m+\frac{3}{2}}{m+\frac{1}{2}}$	$2m+3$	Telescoping
2	$T^{(m)}$	$\frac{1}{2}$	$-\frac{1}{2}$	1	1	$\dfrac{m+\frac{1}{2}}{m-\frac{1}{2}}$	$2m+1$	Telescoping
3	$S^{(m+1)}$	1	$\frac{1}{2}$	$\frac{1}{2}$	$\dfrac{4}{\pi}$	$\dfrac{m+1}{m+\frac{1}{2}}$	$\dfrac{2\,\Gamma(m+2)}{\sqrt{\pi}\,\Gamma(m+\frac{3}{2})}$	S-shift
4	$S^{(m+\frac{3}{2})}$	$\frac{3}{2}$	1	$\frac{1}{2}$	$\dfrac{3}{2}$	$\dfrac{m+\frac{3}{2}}{m+1}$	$\dfrac{2\,\Gamma(m+\frac{5}{2})}{\sqrt{\pi}\,\Gamma(m+2)}$	S-shift

Experiment log (2026-04-09): Fractional β sweep over reduced p/q with $1 \le q \le 12$, $|\beta| \le 3$, and $\gcd(p,q) = 1$ (266 cases). The best cubic candidates reached 81.1+ stable digits, but no PSLQ relation was found in the rational, $\Gamma(1/3)$, or $\Gamma(2/3)$ sectors; the $\Gamma(1/4)$ and $\Gamma(3/4)$ sectors remain under-resolved.

4 The Dichotomy Principle

The central structural result is:

Theorem 2 (Dichotomy Principle). *Let $\{V(m)\}_{m\ge 0}$ be a PCF family of the form* (3) *whose consecutive-member ratio satisfies*

$$\frac{V(m)}{V(m-1)} = \frac{m+\alpha}{m+\beta}$$

for constants α, β with $\alpha \ne \beta$. Then:

(T) *If $\alpha - \beta \in \mathbb{Z}$, then $V(m) \in \mathbb{Q}$ for all $m \in \mathbb{Z}_{\ge 0}$ (provided $V(0) \in \mathbb{Q}$).*

(S) *If $\alpha - \beta \in \mathbb{Z} + \frac{1}{2}$, then $V(m)$ lies in $\mathbb{Q} \cdot \pi^{-1}$ for m of one parity and in $\mathbb{Q}$ for m of the other, with the transcendental factor $\sqrt{\pi}$ present in* (5).

Proof sketch. From (5), the m-dependent factor is the ratio

$$R(m) \;=\; \frac{\Gamma(m+\alpha+1)}{\Gamma(m+\beta+1)}\,.$$

Case (T): When $d := \alpha - \beta \in \mathbb{Z}_{>0}$ (the case $d < 0$ is analogous), the Gamma ratio reduces to a Pochhammer symbol:

$$R(m) = (m+\beta+1)_d = \prod_{j=0}^{d-1}(m+\beta+1+j)\,,$$

which is a polynomial in m with rational coefficients (since $\beta \in \frac{1}{2}\mathbb{Z}$). The prefactor $\Gamma(\beta+1)/\Gamma(\alpha+1)$ is likewise rational when d is a positive integer, because $\Gamma(\alpha+1)/\Gamma(\beta+1)$ is itself a Pochhammer with rational argument. Hence $V(m) = V(0) \cdot [\text{rational}]$.

Concretely, for Family 1 ($\alpha = 3/2$, $\beta = 1/2$, $d = 1$):

$$R(m) = m + \tfrac{3}{2}, \quad \frac{\Gamma(3/2)}{\Gamma(5/2)} = \frac{2}{3}, \quad V(m) = 3 \cdot \tfrac{2}{3}(m + \tfrac{3}{2}) = 2m + 3.$$

Case (S): When $\alpha - \beta = k + \frac{1}{2}$ with $k \in \mathbb{Z}$, the Gamma ratio $R(m)$ still reduces to a Pochhammer, but the prefactor $\Gamma(\beta+1)/\Gamma(\alpha+1)$ involves Γ at half-integer vs. integer arguments. By the Legendre duplication formula,

$$\Gamma\big(n + \tfrac{1}{2}\big) = \frac{(2n)!}{4^n\, n!}\sqrt{\pi}\,,$$

so the ratio acquires an ineliminable factor of $\sqrt{\pi}$ (or $1/\sqrt{\pi}$). This factor propagates through (5); at integer m the Γ-terms in $R(m)$ alternate between producing an additional $\sqrt{\pi}$ and cancelling it, giving the parity pattern $V(m) \in \mathbb{Q}$ versus $V(m) \in \mathbb{Q}/\pi$.

Concretely, for Family 4 ($\alpha = 3/2$, $\beta = 1$, $d = 1/2$):

$$V_4(m) = \frac{2\,\Gamma(m + \frac{5}{2})}{\sqrt{\pi}\,\Gamma(m+2)}\,,$$

and PSLQ confirms $V_4 \cdot \sqrt{\pi} \cdot \Gamma(m+2)/\Gamma(m+\frac{5}{2}) = 2$ to 200 digits for all tested m values [6]. □

Corollary 3. *Evaluating the S-type families at half-integer m swaps the transcendental content: Family 4 yields $V_4(-\frac{1}{2}) = 4/\pi$, $V_4(\frac{1}{2}) = 16/(3\pi)$, and $V_4(\frac{3}{2}) = 32/(5\pi)$, while integer m gives $V_4(0) = 3/2$, $V_4(1) = 15/8$, $V_4(2) = 35/16$.*

The phase diagram in Figure 1 visualises the dichotomy: integer values of the gap $\alpha - \beta$ correspond to rational (T-type) families, while half-integer gaps produce transcendental (S-type) families.

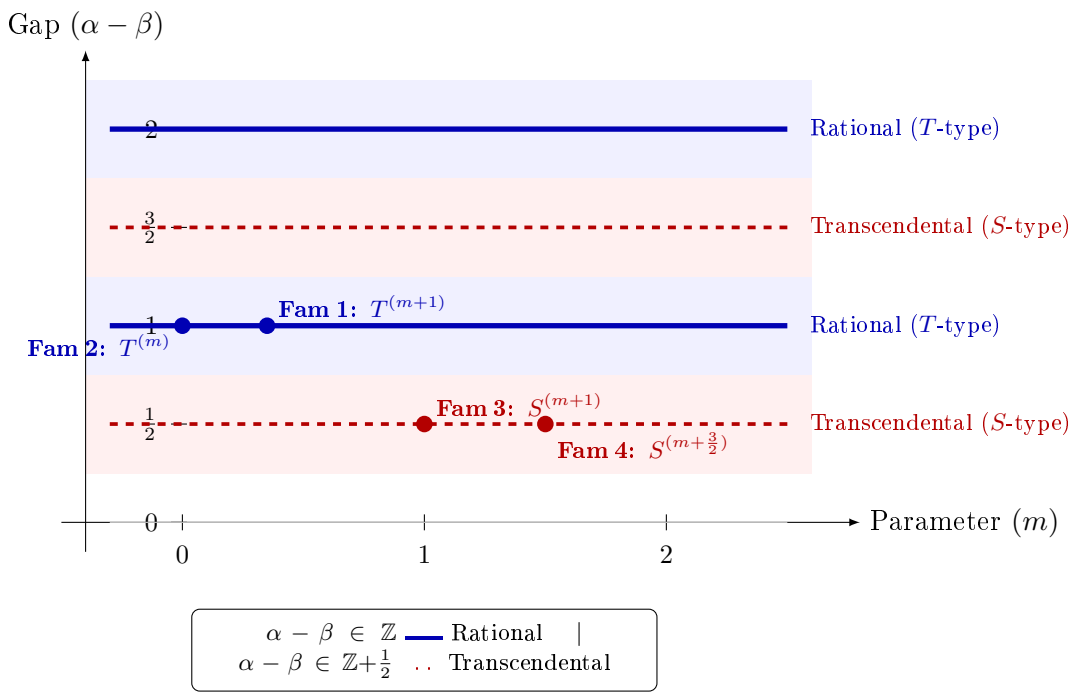

Figure 1: Phase diagram of PCF convergence types in the $(\alpha-\beta, m)$ plane. Rational solutions coalesce on integer gaps ($\alpha - \beta \in \mathbb{Z}$, solid blue), while transcendental values emerge at half-integer shifts ($\alpha - \beta \in \mathbb{Z} + \frac{1}{2}$, dashed red). Shaded bands indicate the predicted continuation of each zone to higher gap values.

5 Connection to the S-family

Families 3 and 4 are shifts of the classical S-family defined in (2). Specifically:

- Family 3 satisfies $V_3(m) = S^{(m+1)}$, an integer shift. Its a_n polynomial can be rewritten as $a_n = -n(2n - 2(m+1) - 1)$, which is the S-family at parameter $m' = m + 1$.
- Family 4 satisfies $V_4(m) = S^{(m+3/2)}$, a half-integer shift. Here $a_n = -n(2n - 2(m+\frac{3}{2}) - 1)$, which is the S-family at parameter $m' = m + 3/2$.

Both identities are confirmed by direct PCF-vs-PCF comparison to 550 digits via the evaluation code accompanying this paper.

Families 1 and 2, by contrast, are *not* shifts of the S-family: they use $b_0 = 2$ and $b_n = 3n+2$ (rather than $b_0 = 1$, $b_n = 3n + 1$), and the negative m-coupling ($\sigma = -1$) produces the integer gap $\alpha - \beta = 1$ that triggers telescoping. We denote these the T-*type* families.

6 Concluding remarks

6.1 Open cubic PCF constant from the fractional-β scan

Table 2: Master summary table for currently open higher-degree PCF cases.

(d, p, α, β)	Value	Basis size	Verdict
$(3, 1, 6, 5/3)$	$1.78856\ldots$	24	OPEN

6.2 Degree-3 and degree-4 numerators: a refined picture

The fractional-β scan tests the generalized Dichotomy Principle for $d \in \{3, 4\}$ at 150 decimal digits and truncation depths 200, 350, and 500 by choosing β so that $\rho = \alpha - \beta$ has prescribed fractional part. Table 3 records the targeted cases. The reported digit count is the minimum agreement across the depth comparisons $(200, 350)$, $(350, 500)$, and $(200, 500)$. PSLQ was applied only when this stability exceeded 15 digits, with coefficient bound 500 in the requested rational or $\Gamma(j/d)$ bases.

Table 3: Fractional-β scan for $d \in \{3, 4\}$, computed at dps = 150 and depth 500.

(d, p, α, β)	ρ	ρ mod 1	result	digits
$(3, 1, 4, 1)$	3	0	no rational hit	43.91
$(3, 1, 5, 2)$	3	0	no rational hit	54.57
$(3, 2, 6, 3)$	3	0	no rational hit	46.87
$(3, 1, 4, 5/3)$	$\frac{7}{3}$	$\frac{1}{3}$	no $\Gamma(1/3), \Gamma(2/3)$ hit	44.29
$(3, 1, 5, 5/3)$	$\frac{10}{3}$	$\frac{1}{3}$	no $\Gamma(1/3), \Gamma(2/3)$ hit	54.40
$(3, 1, 6, 5/3)$	$\frac{13}{3}$	$\frac{1}{3}$	no $\Gamma(1/3), \Gamma(2/3)$ hit	64.22
$(3, 2, 4, 5/3)$	$\frac{7}{3}$	$\frac{1}{3}$	no $\Gamma(1/3), \Gamma(2/3)$ hit	31.64
$(3, 1, 3, 2/3)$	$\frac{7}{3}$	$\frac{1}{3}$	no $\Gamma(1/3), \Gamma(2/3)$ hit	33.27
$(3, 1, 4, 1/3)$	$\frac{11}{3}$	$\frac{2}{3}$	no $\Gamma(1/3), \Gamma(2/3)$ hit	43.50
$(3, 1, 5, 2/3)$	$\frac{13}{3}$	$\frac{1}{3}$	no $\Gamma(1/3), \Gamma(2/3)$ hit	53.86
$(4, 1, 5, 3/4)$	$\frac{17}{4}$	$\frac{1}{4}$	not tested	10.63
$(4, 1, 6, 3/4)$	$\frac{21}{4}$	$\frac{1}{4}$	not tested	12.54
$(4, 1, 5, 7/4)$	$\frac{13}{4}$	$\frac{1}{4}$	not tested	10.92
$(4, 1, 5, 1/4)$	$\frac{19}{4}$	$\frac{3}{4}$	not tested	10.47
$(4, 1, 6, 1/4)$	$\frac{23}{4}$	$\frac{3}{4}$	not tested	12.39

The rational-control cases ($\rho \in \mathbb{Z}$) all converge stably and exhibit no unexpected Γ-sector relation. The cubic cases with $\rho \bmod 1 \in \{1/3, 2/3\}$ also converge stably, but they resist identification in the standard $\Gamma(1/3), \Gamma(2/3)$ basis at coefficient bound 500. For the strongest cubic example $(p, \alpha, \beta) = (1, 6, 5/3)$ with $d = 3$, the value at depth 600 is approximately

$1.7885654281636925083376\ldots$, `mp.identify` returns no recognition, its simple continued fraction begins $[1; 1, 3, 1, 2, 1, \ldots]$, and an extended PSLQ search including $\pi^{1/3}, \pi^{2/3}, 3^{1/3}$, and $\log 3$ combinations with $\Gamma(1/3), \Gamma(2/3)$ still finds no relation at coefficient bound 500.

Remark 4. These computations do not confirm the naive residue-class extension of the Dichotomy Principle to $d \geq 3$. At the same time, they do not constitute a formal refutation of every possible $\Gamma(k/d)$ description, since PSLQ only tests the specified low-coefficient bases. The evidence instead suggests that any correct generalization for $d \geq 3$ must involve additional structural conditions beyond the characteristic exponent difference ρ, and possibly a richer period basis incorporating terms such as $\pi^{1/d}$ or $d^{1/d}$.

The quartic 1/4 and 3/4 buckets remain under-resolved at the tested depth, so the data refine the question rather than close it.

6.3 Fractional β sweep: negative result for $q \leq 12$

A dense follow-up sweep tested reduced fractions $\beta = p/q$ with $1 \leq q \leq 12$, $|\beta| \leq 3$, and $\gcd(p, q) = 1$, giving 266 targeted cases across the cubic and quartic candidates singled out by the earlier dichotomy search. The strongest cubic cases, $(3, 1, 6, 8/3)$ and $(3, 1, 6, 7/3)$, reached 81.1+ and 80.9+ stable digits respectively, so the negative outcome is not attributable to a lack of numerical stability at the current depth settings.

Despite this, no relation was detected in the rational, $\Gamma(1/3)$, or $\Gamma(2/3)$ families at the 15-digit PSLQ threshold. This sweep therefore strengthens the evidence that the verified Rosetta lines are concentrated on the integer and half-integer regimes already represented in Table 1. What remains open is narrower: the $\Gamma(1/4)$ and $\Gamma(3/4)$ buckets are still under-resolved (they remained below 15 stable digits), and genuinely new behaviour could still emerge for finer fractional parameters with $q > 12$.

The Dichotomy Principle provides a clean arithmetic criterion for the nature of PCF limits within the quadratic–linear class studied here. Several directions remain:

- A fully rigorous proof (beyond the sketch in Theorem 2) via Euler's continued fraction theorem and Gauss summation of ${}_2F_1(a, b; c; 1)$.
- Extension to cubic a_n or quadratic b_n, where the ratio structure may involve ${}_3F_2$ hypergeometric functions.
- Lean 4 formalisation of the telescoping identities.

All computational verification code (PCF evaluators, PSLQ pipelines, and the systematic search engine) is available at the repository accompanying this manuscript.

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